

- Rare view into iron-ore melting furnace with inflow pellets during inspection cycle (Corus).



Supercritical shallow granular and slurry flows through a contraction

Onno Bokhove

o.bokhove@math.utwente.nl

Vreman et al., JFM 2007

Akers & B., Physics of Fluids 2008

Rhebergen & B., Van der Vegt, Comp. Meth. Appl. Mech. Eng. 2009

Numerical Analysis and Computational Mechanics Group,

Dept. of Applied Mathematics, University of Twente, The Netherlands

Institute of Mechanics, Processes and Control, Twente

GFD Program Woods Hole Oceanographic Institution

Outline

- 1. Introduction
- 2. Experiments: *granular*, *hydraulic* & *slurry*
- 3. Multiple steady states: 1D theory, *granular* & *hydraulic*
- 4. Supercritical flow: 2D theory
- 5. Shallow slurry flows: 2D simulations
- 6. Conclusions
- *References*

1. Introduction

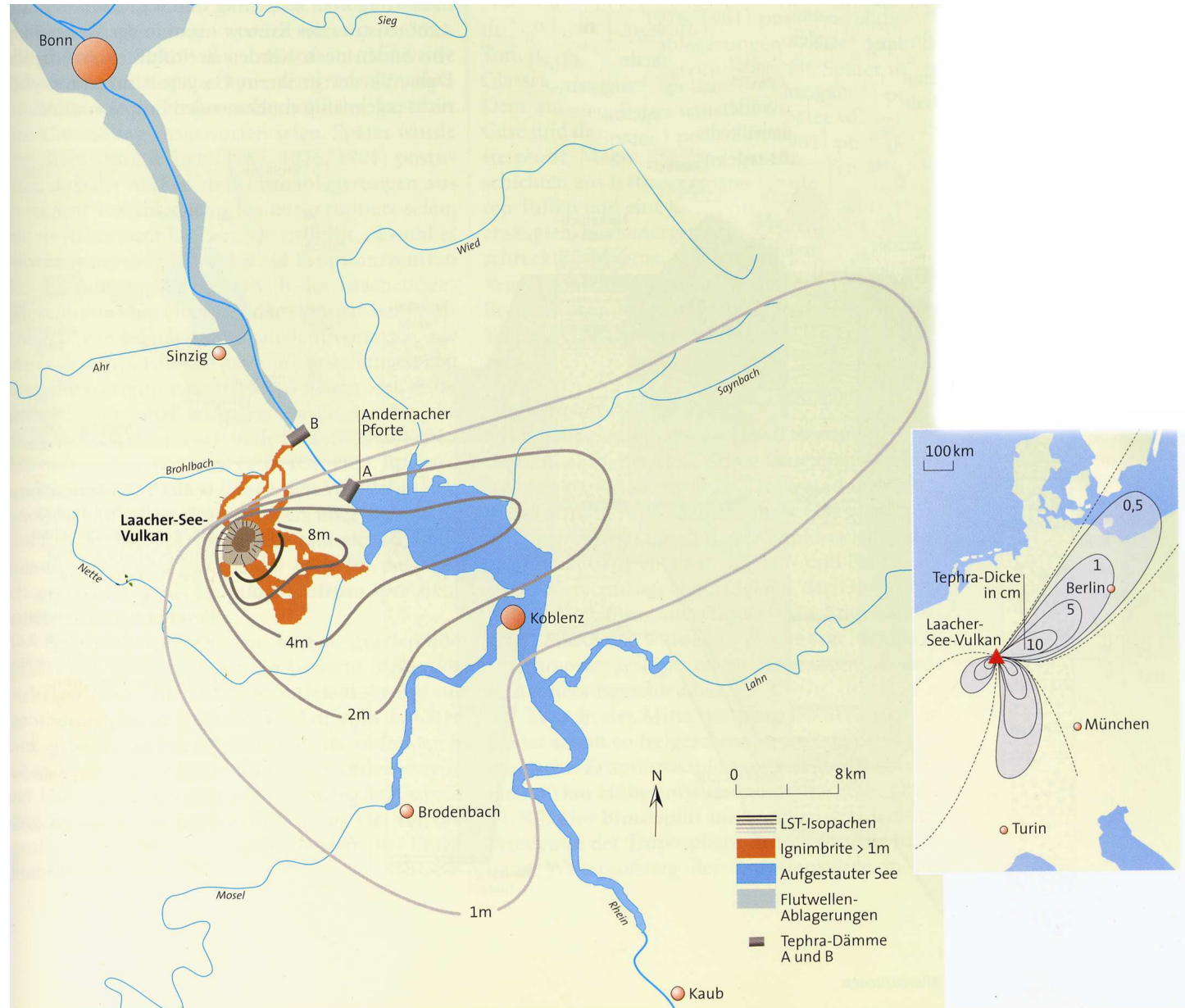
Once upon a time, ... a field trip in the *German Eifel* (2002):

- Under guidance of Hans-Ulrich Schmincke.
- Eifel is “young” volcanological field 2.5hrs from Twente.
- **Laacher See Volcano** exploded in Pleistocene (12.9k aBP):



Laacher See Volcano

- 1–8 meter
layer of
tephra
around
volcano
causing
fluidized
tephra
flows
- Schmincke
2000



... Introduction

Research question:

- Can we make a theory and laboratory experiment in support of this event of dam and lake formation and its collapse?
- Complexity of carrier fluid (“the Rhine”) and floating granular material suggests initial simplification:
- *What flow regimes emerge when dry granular matter flows down an inclined chute with a contraction?*

... Introduction

- We will consider shallow *granular*, *hydraulic* & *slurry* flows through a contraction, experimentally, analytically and numerically.
- Large variations in particle/water discharges through contracting channels can lead to dramatic changes in the flow state.
- *Surges: mud and river flows underneath bridges and in ravines.*
- *Linear contraction: archetypical contraction geometry.*
- E.g. contraction flow with two oblique hydraulic jumps:

... Introduction



Fig. 1. H_2O : Oosterschelde storm surge barrier.

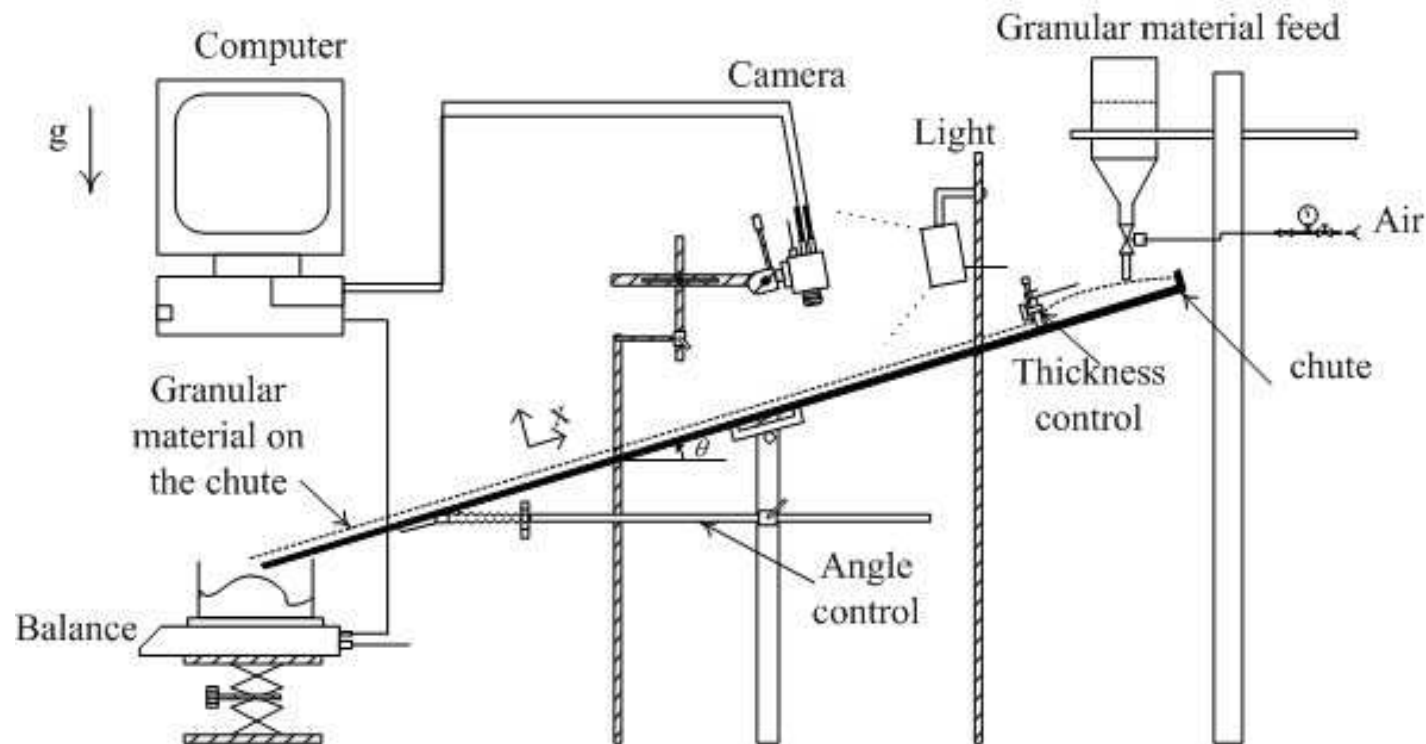
... Motivation

- How do *granular*, *hydraulic* and *slurry* flows through a contraction compare with another, on inclined and flat chutes?
- Can we experimentally **find three stable steady states** for one discharge rate?
 - Baines & Whitehead (2003) couldn't, for flow over hill.
- Industrial counterparts: proposal Twente & Eindhoven, rotating granular flows (Corus, BASF, Unilever).

2. Experiments

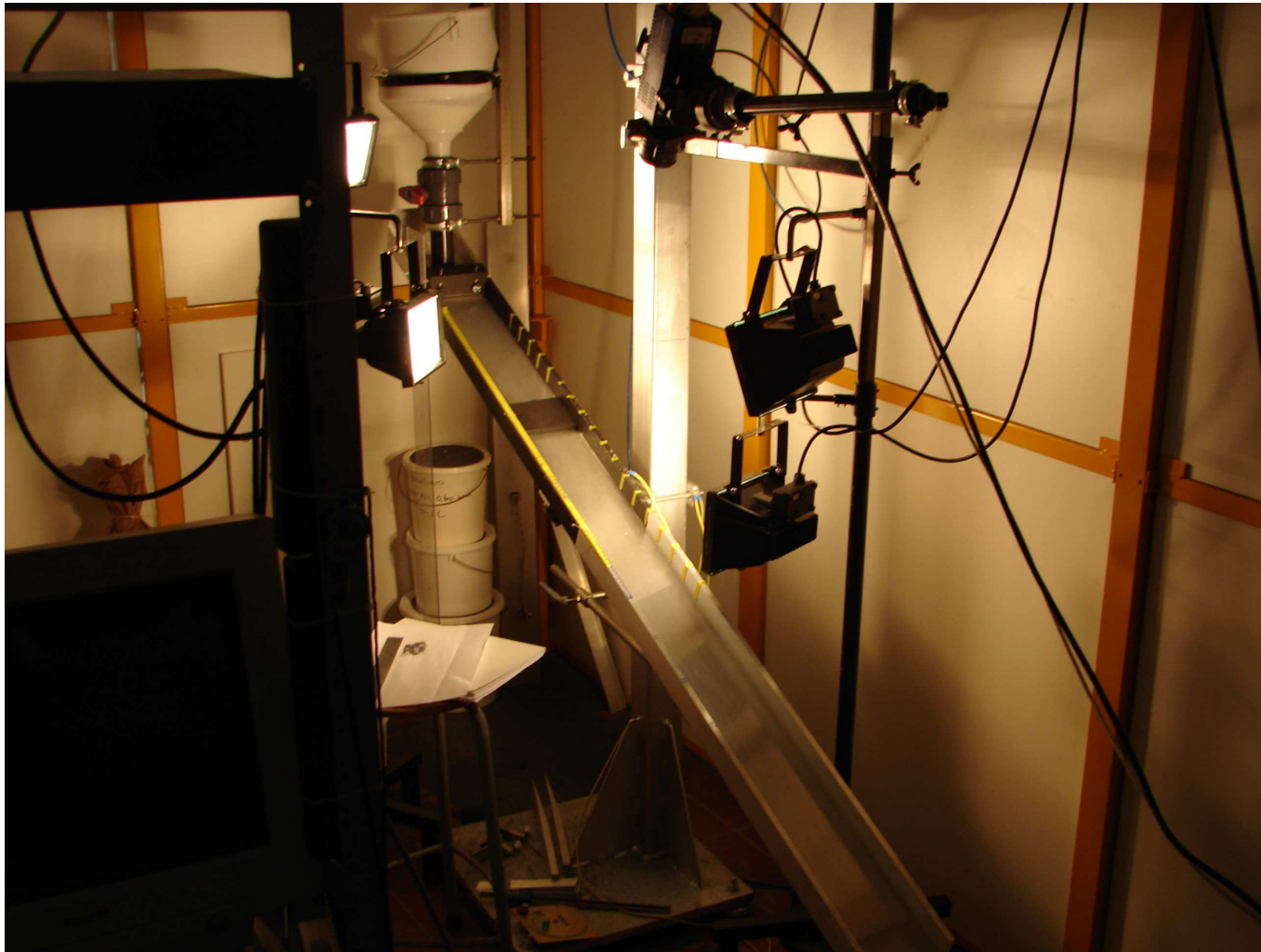
Granular flow (Al-Tarazi, Vreman) Inclined aluminum chute:

- $b_0 = 13\text{cm}$ wide, 2m long, inclination θ , $\phi = 15^\circ - 35^\circ$
- **glass beads: inelastic**, diameter: S0) $d \in [.5, .6]\text{mm}$, S) $[.28, .42]\text{mm}$, M) $[.4, .6]\text{mm}$, L) $[.75, 1]\text{mm}$; density: 2470kg/m^3



... Granular experiments

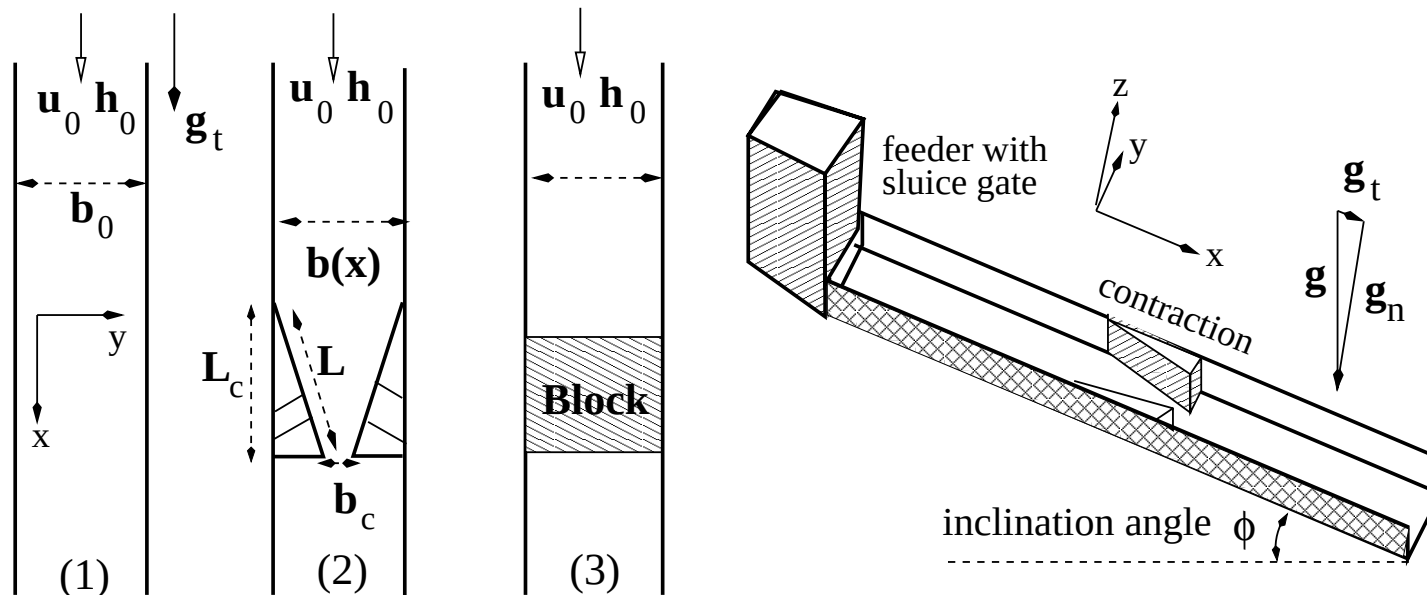
- Granular flow. Set-up: impression



... Granular experiments

Top view sketches of granular experiments on **inclined chute**:

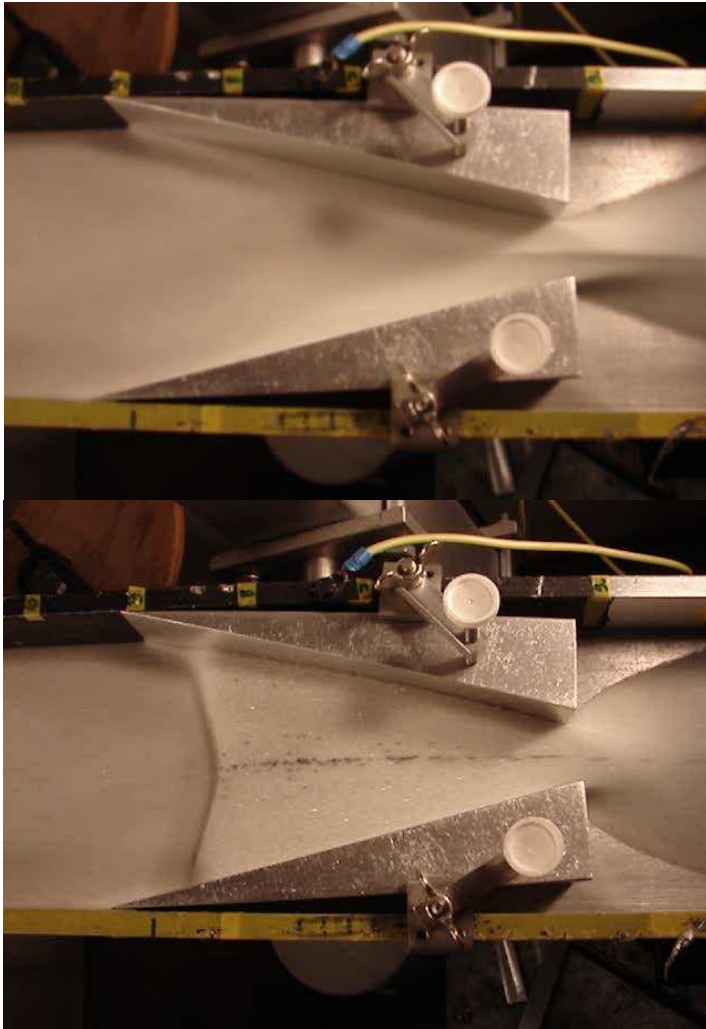
- [1] of constant width with no contraction, $b_c = b_0$,
- [2] with a **localized contraction**, and
- [3] blocked in middle, " $b_c = 0$ ".



Reason: establish **upstream conditions** & bores experimentally.

Dry *granular* flow on inclined chute:

- *Transition* from supercritical to granular jump state:



- Steady-state *granular* flows: parameter plane upstream Froude # $F_0 = u_0/\sqrt{g h_0}$ & scaled minimum nozzle width $B_c = b_c/b_0$:

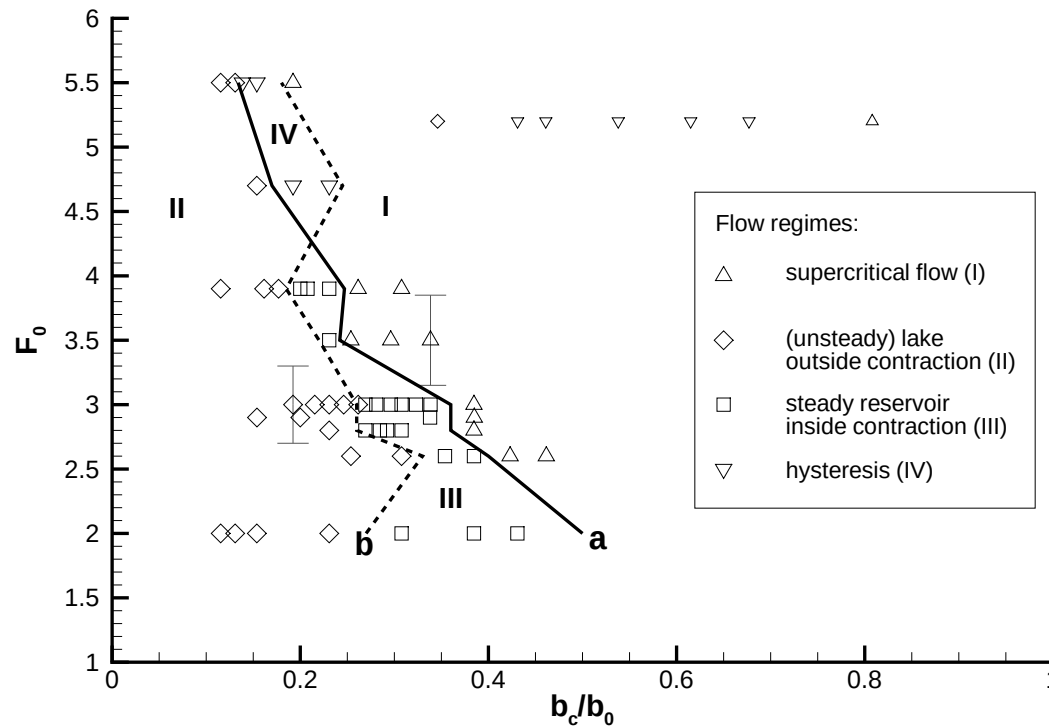


Fig. 2. $\triangle$: supercritical oblique jumps; ∇ : hysteretic oblique jumps;
 $\square$: steady reservoir; $\diamond$: lake & upstream bore

Hydraulic/slurry ... experiments

- Horizontal flume: 110cm long, $b_0 = 20\text{cm}$ wide, with sluice gate:

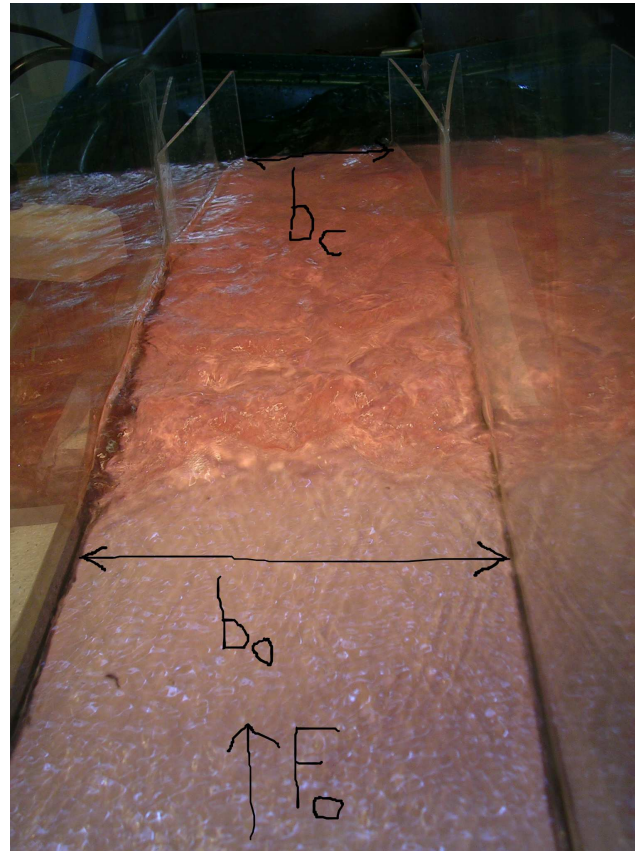


Fig. 3. Looking at the contraction in the flume; $F_0 = u_0/\sqrt{g h_0}$, $B_c = b_c/b_0$.

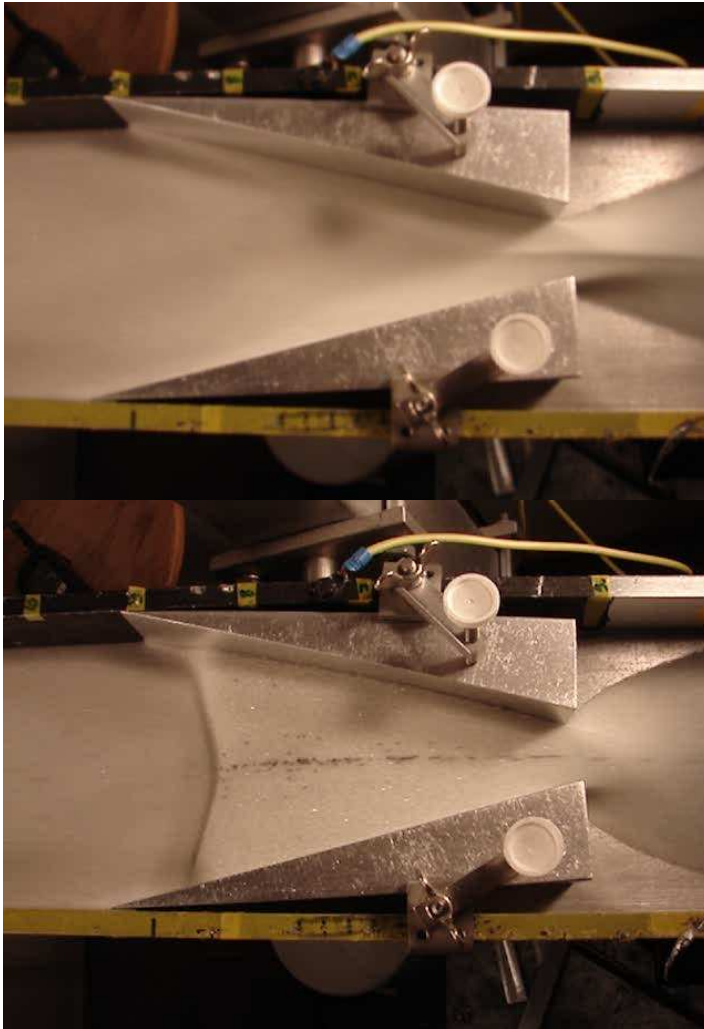
- Three multiple steady states in regions (i/iii/iv):



Fig. 3. Multiple states: $F_0 = 3.07$ & $B_c = 0.7$. Left to right: upstream steady shock (iii), reservoir state with “Mach stem” (iv), and oblique waves (i). Transitions induced by pushing flow.

Compare with dry **granular** flow on inclined chute:

- **Transition** from supercritical to granular jump state:



- Steady-state *hydraulic* flows: parameter plane of upstream Froude number F_0 and scaled minimum nozzle width $B_c = b_c/b_0$:

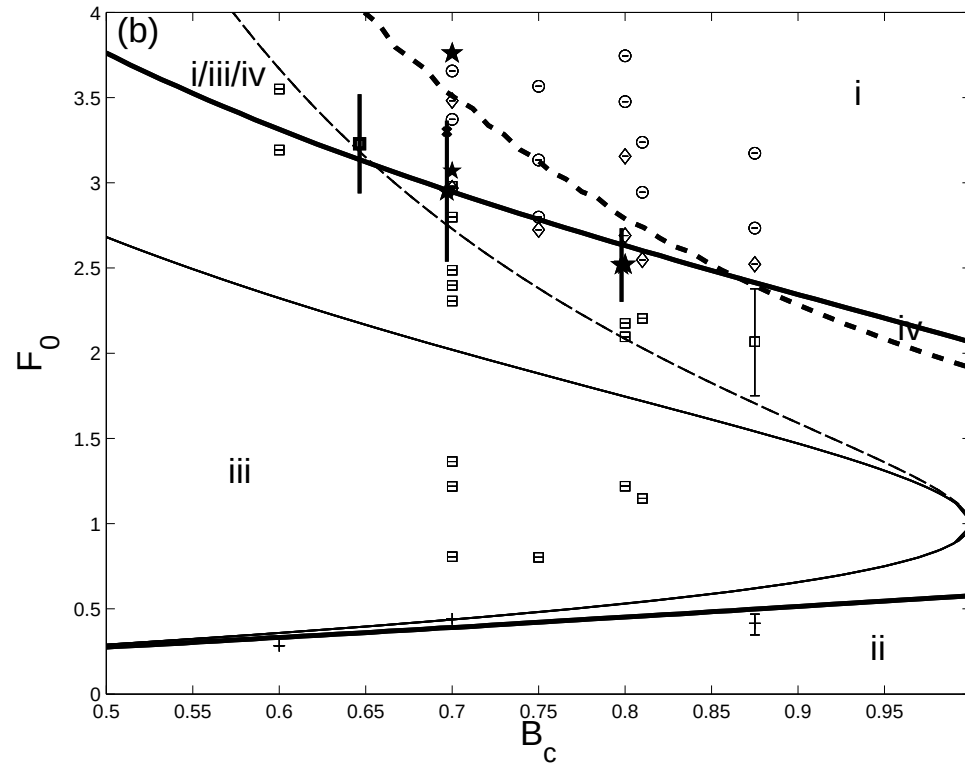


Fig. 4. +: smooth flows (ii); □: upstream moving shocks (iii); ◇: steady shocks (iii); o: oblique waves (i). ★: flows with 3 possible states.

- Stars, three multiple steady states in regions (i/iii/iv):



Fig. 4'. Multiple states: $F_0 = 3.07$ & $B_c = 0.7$. Left to right: upstream steady shock (iii), reservoir state with “Mach stem” (iv), and oblique waves (i). Transitions induced by pushing flow.

- Transition induced upstream *slurry* avalanche polystyrene beads:

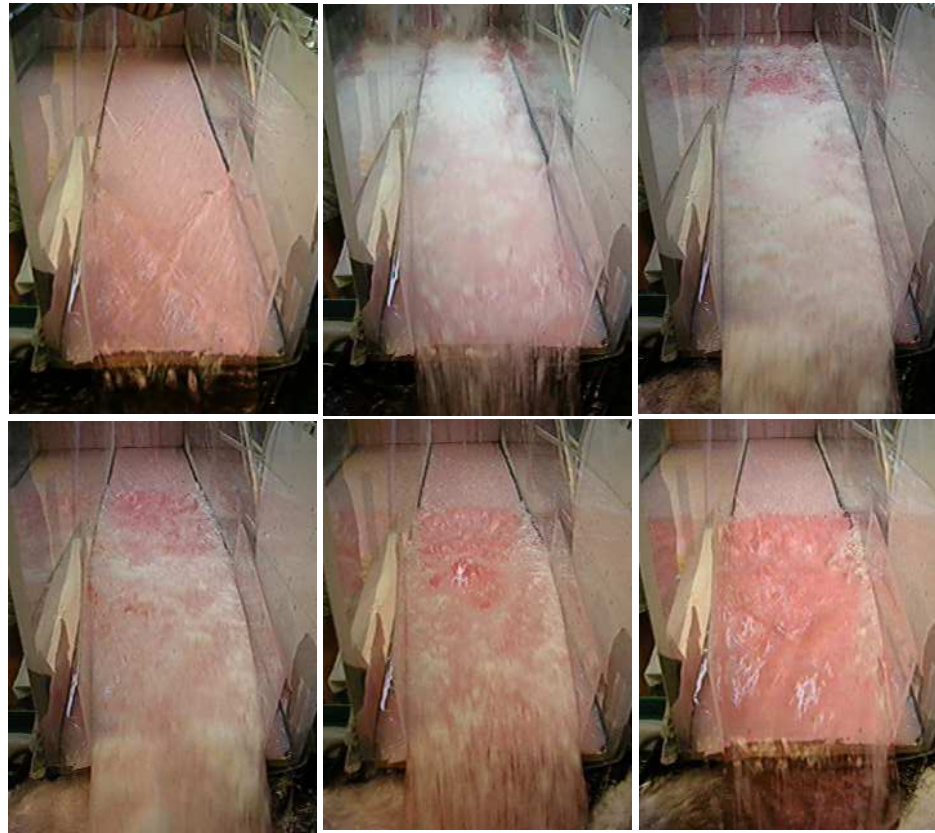


Fig. 5. Oblique wave state (top left) to upstream steady shock state (bottom right). $t=1$ to $6s$. $\sim 900 \text{ kg/m}^3$, $F_0 = 3.07$ & $B_c = 0.7$.

3. Multiple steady states: 1D theory

- Steady-state and “shock” solutions of dimensionless 1D cross-sectionally averaged shallow water equations:

$$u_t + u u_x + h_x / F_l^2 = \begin{cases} \sim (\tan \phi - \mu(F)) & \text{granular} \\ -C_d u^2 / h & H_2O \end{cases} \quad (1)$$

$$(bh)_t + (buh)_x = 0 \quad (2)$$

- with velocity $u = u(x, t)$, depth $h = h(x, t)$, width $b = b(x)$,
- Froude number $F_l = F_0$ upstream or $F_l = F_m$ at contraction,
- granular**: angle ϕ , $\mu(F > 1) \leq \tan \phi$, $\mu_{sub}(F < 1) < \tan \phi$
acceleration integral: $Z_{1,2} = \int_{x_0}^{x_c} (\tan \phi - \mu_{sub}) dx / h_0$.
- H₂O**: demarcation lines friction (thin $C_d = 0$, thick: $C_d > 0$).

3. Multiple steady states: 1D theory

- Steady-state and “shock” solutions of dimensionless 1D cross-sectionally averaged shallow water equations:

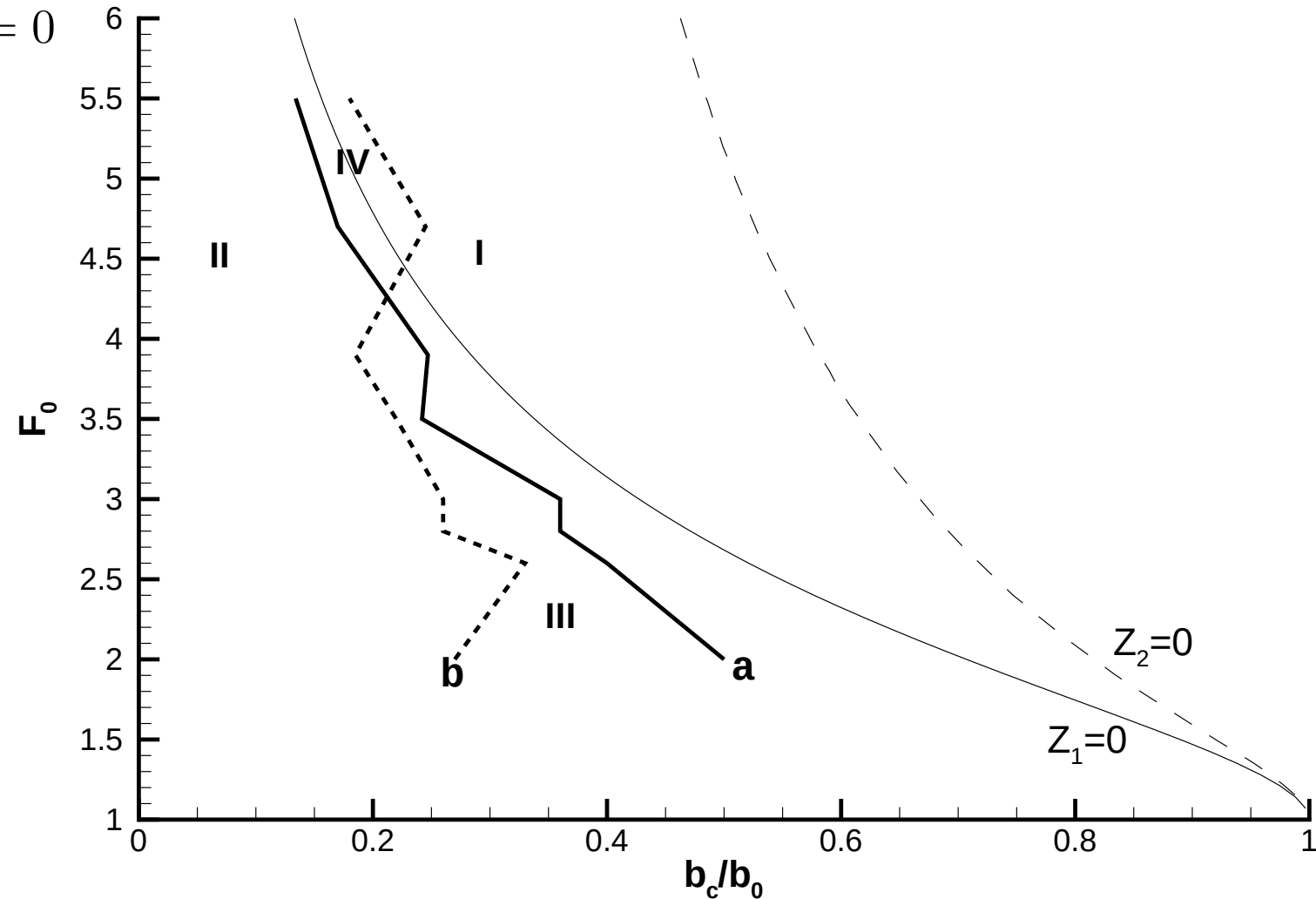
$$\frac{d}{dx}(u^2/2 + h/F_l^2) = \begin{cases} \sim (\tan \phi - \mu(F)) & \text{granular} \\ -C_d u^2/h & H_2O \end{cases} \quad (3)$$

$$buh = Q \quad (4)$$

- with velocity $u = u(x, t)$, depth $h = h(x, t)$, width $b = b(x)$,
- Froude number $F_l = F_0$ upstream or $F_l = F_m$ at contraction,
- granular**: angle ϕ , $\mu(F > 1) \leq \tan \phi$, $\mu_{sub}(F < 1) < \tan \phi$
acceleration integral: $Z_{1,2} = \int_{x_0}^{x_c} (\tan \phi - \mu_{sub}) dx / h_0$.
- H₂O**: demarcation lines friction (thin $C_d = 0$, thick: $C_d > 0$).

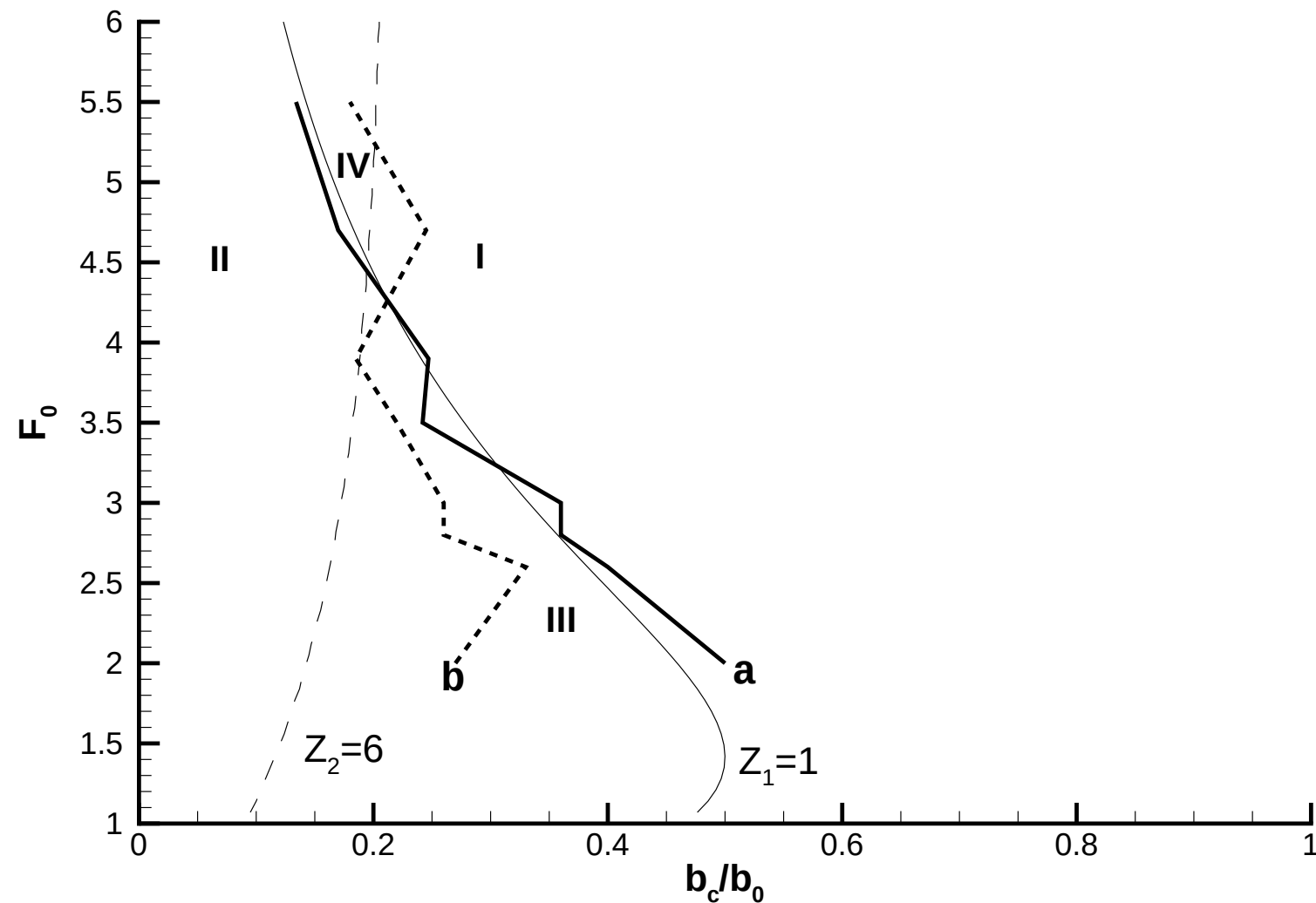
Granular: effectively inviscid F_0, b_c -parameter plane

- Isolines of $Z = 0$
- Solid: critical curve & dashed: shock curve



Granular: modified F_0, b_c -parameter plane best fit Z_1, Z_2

- Isolines of Z
- Solid critical curve & dashed shock curve



- H_2O : steady-state flows: parameter plane F_0 versus $B_c = b_c/b_0$:

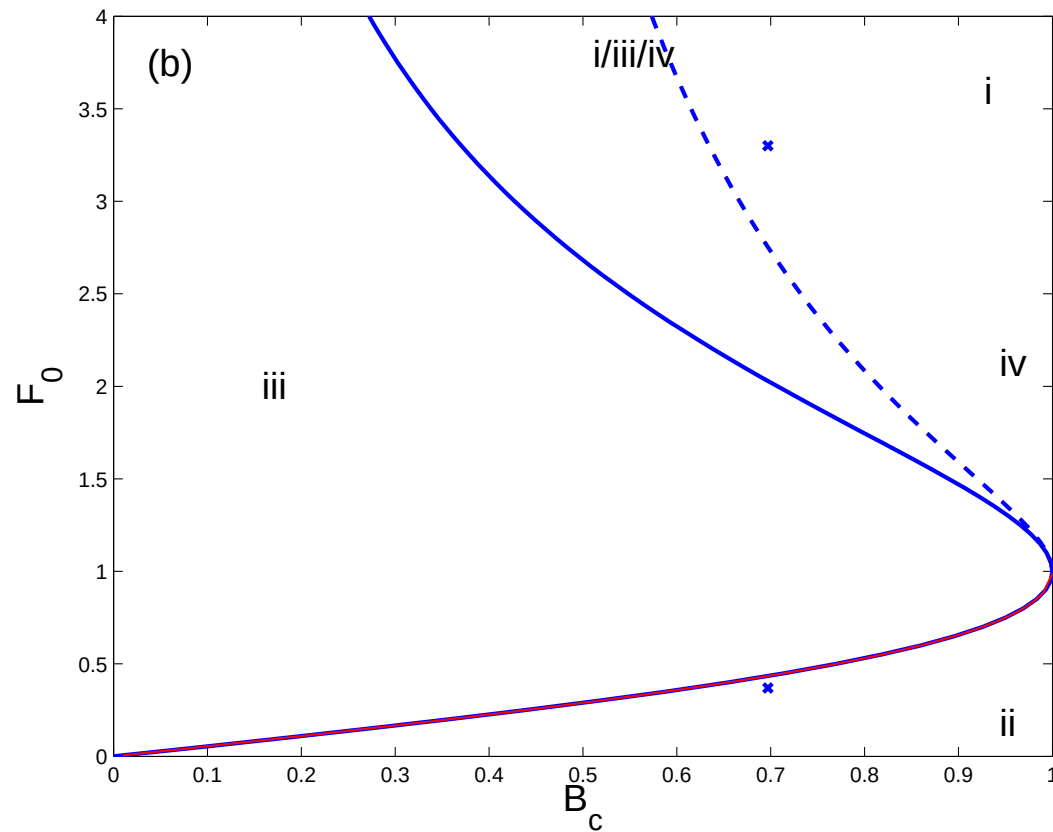


Fig. 4'. From 3 (inviscid)

- Steady-state flows: parameter plane F_0 versus $B_c = b_c/b_0$:

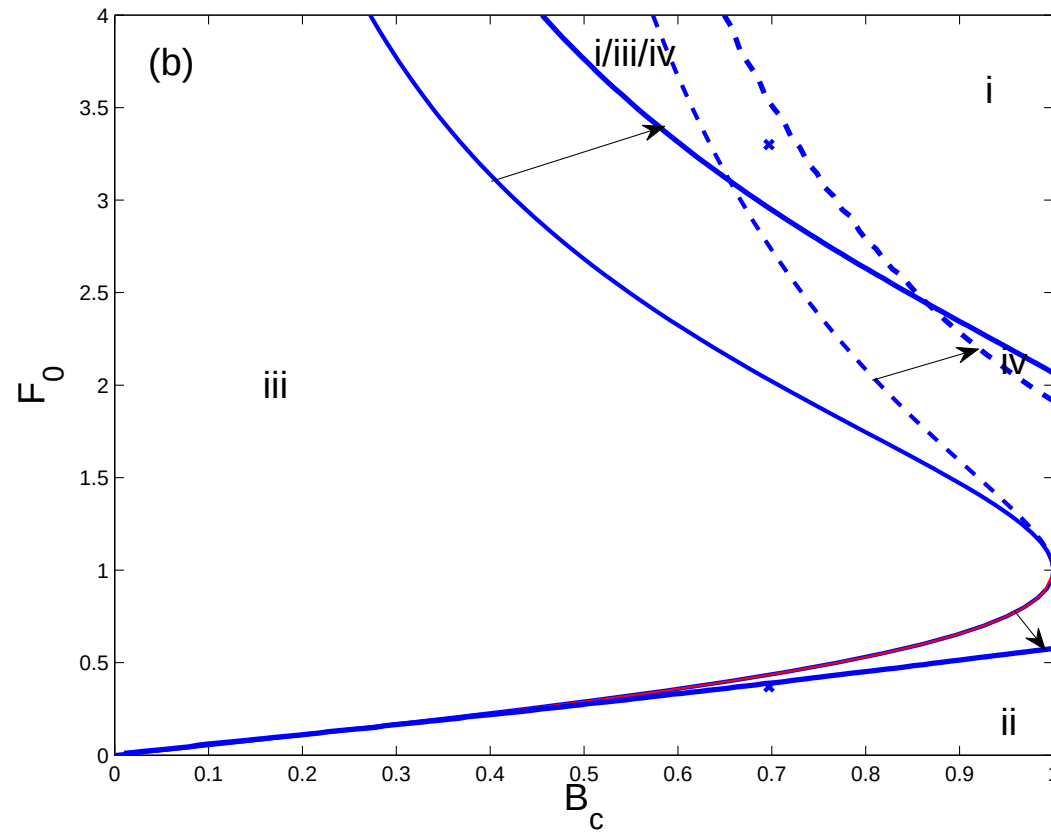
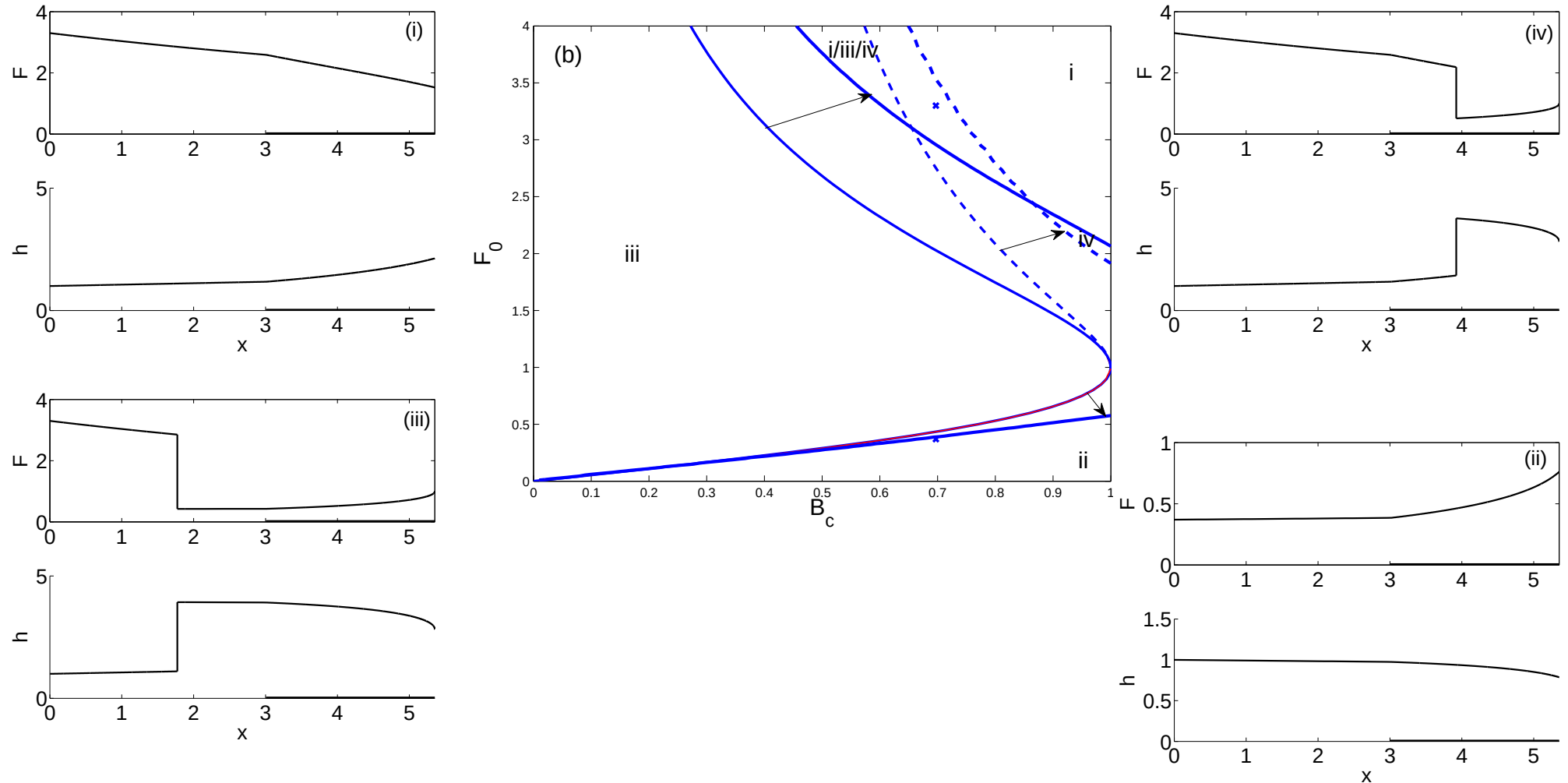
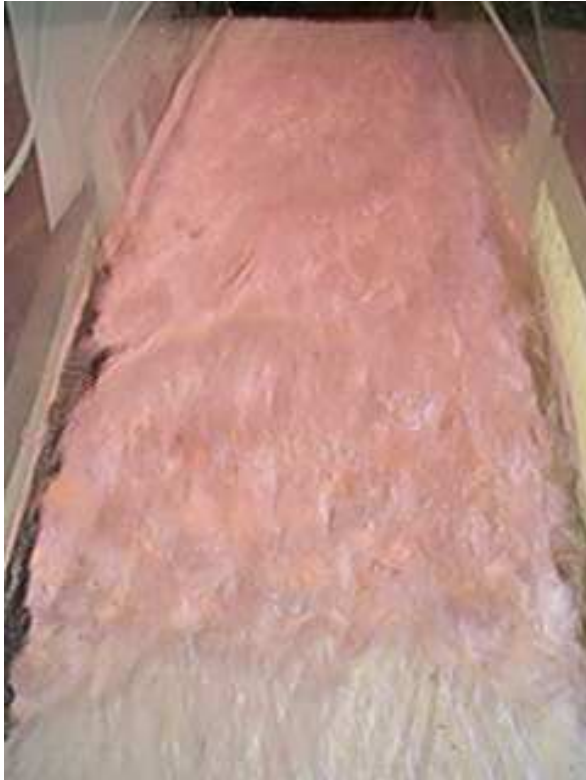


Fig. 4''. ... 3 (inviscid) to 4 (friction) flow regimes: bifurcation.

- Steady-state flows: parameter plane F_0 versus $B_c = b_c/b_0$:



- Three multiple steady states in regions (i/iii/iv):



- Steady-state flows: parameter plane F_0 versus $B_c = b_c/b_0$:

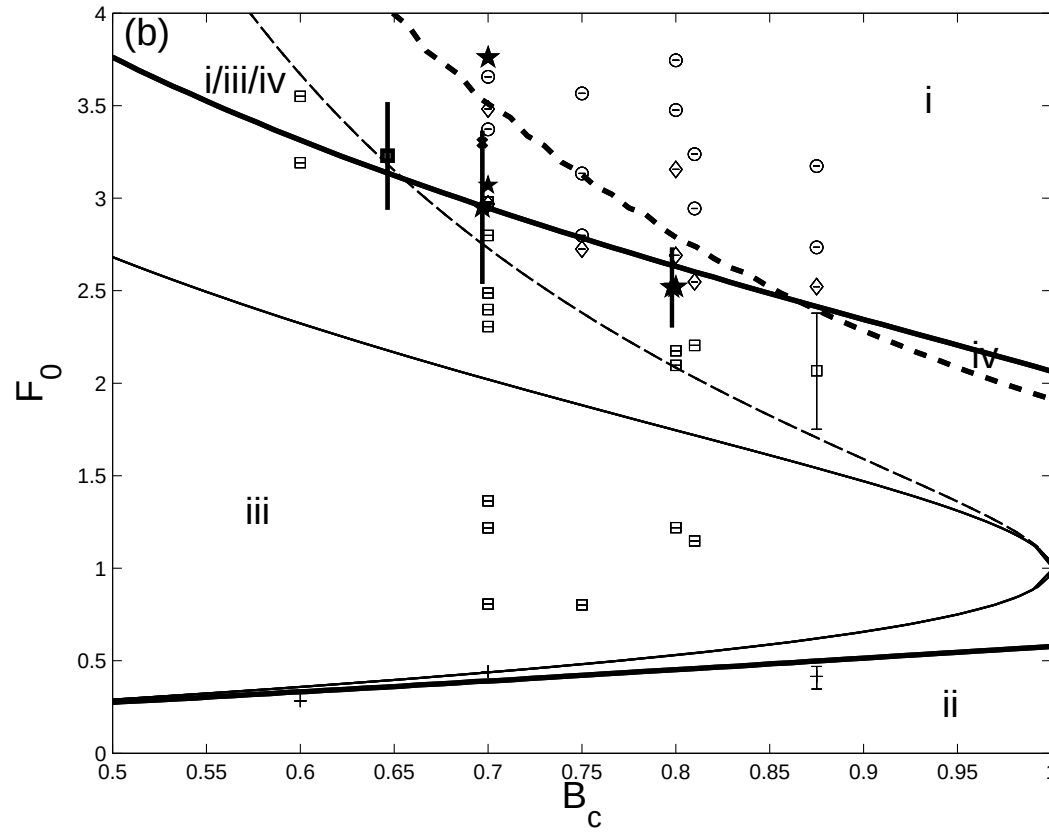


Fig. 4'''. +: smooth flows (ii); □: upstream moving shocks (iii); ◇: steady shocks (iii); ○: oblique waves (i). ★: flows with 3 possible states.

4. Supercritical flows: 2D theory

- Some people did not believe 1D theory.

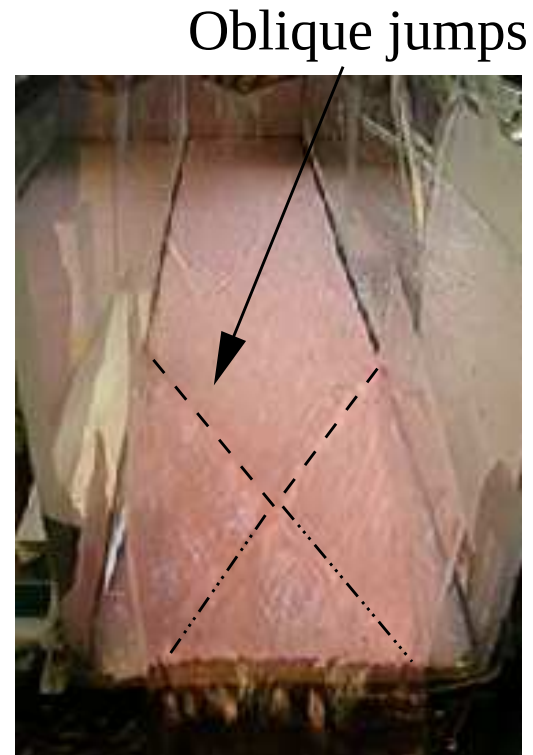
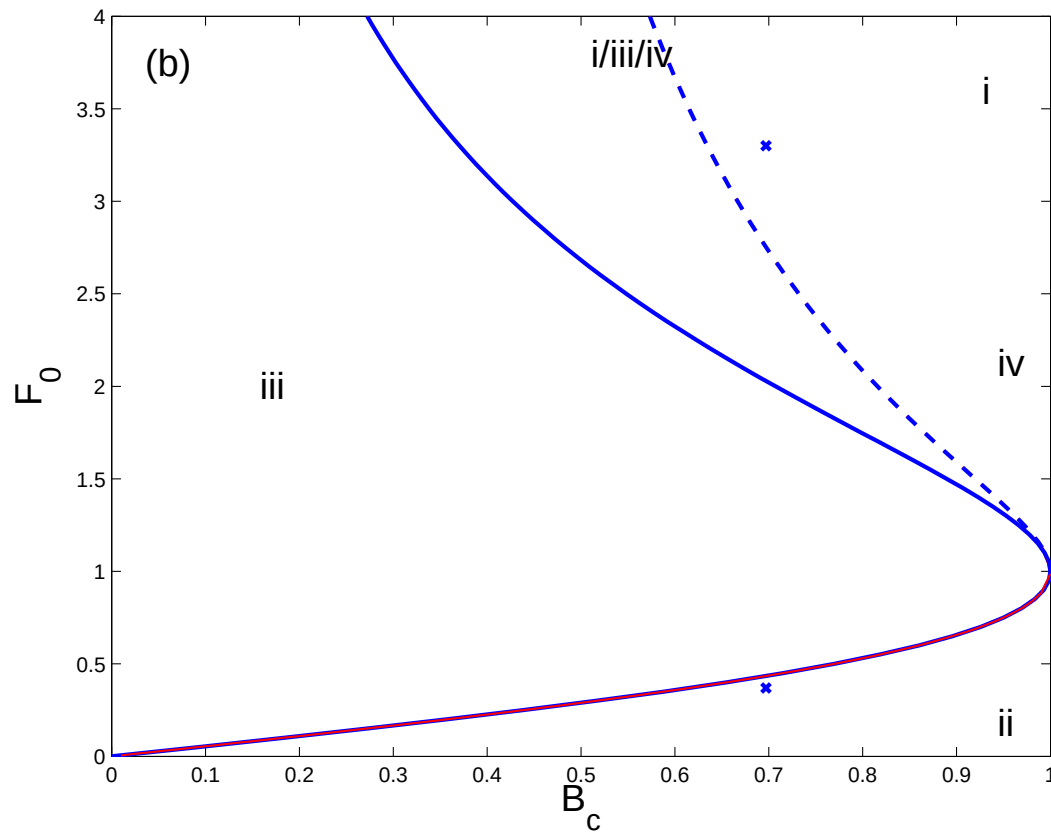


Fig. 4'. ... inviscid case.

...2D theory

- Existence 2D oblique granular/hydraulic jumps no net friction:

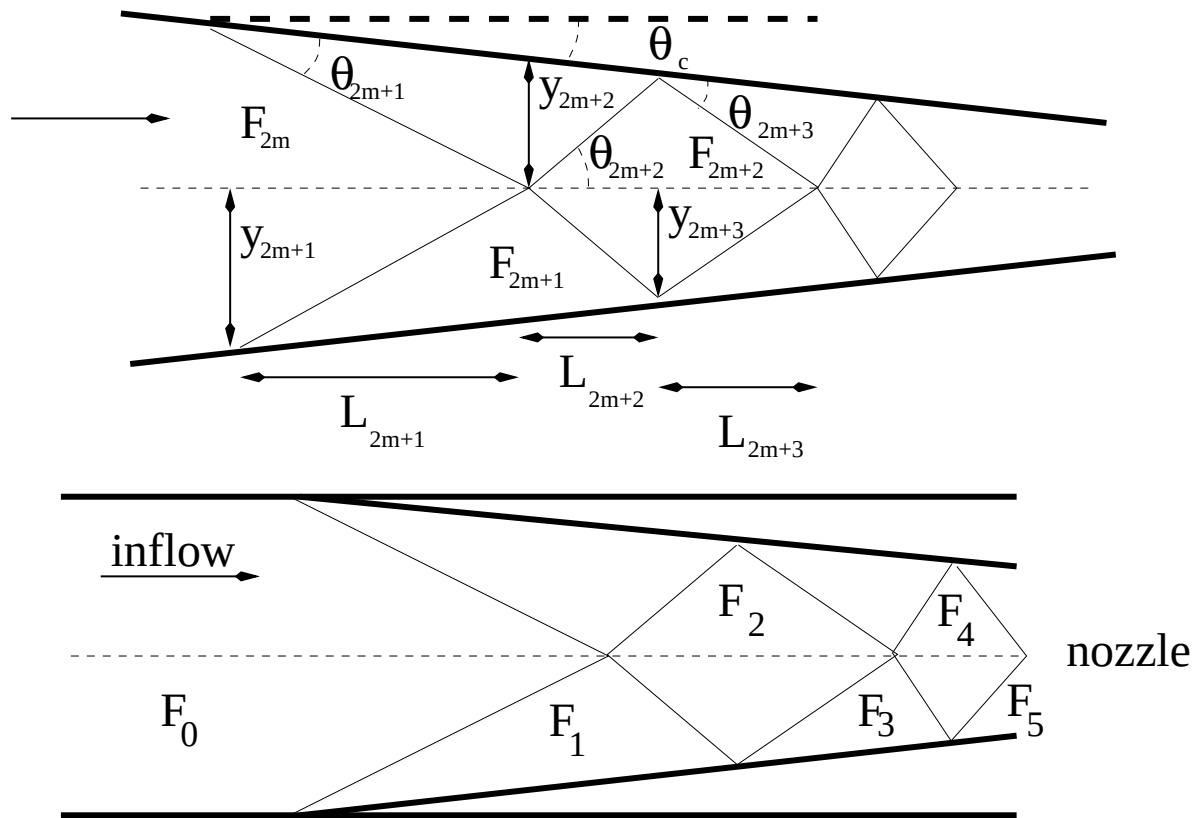
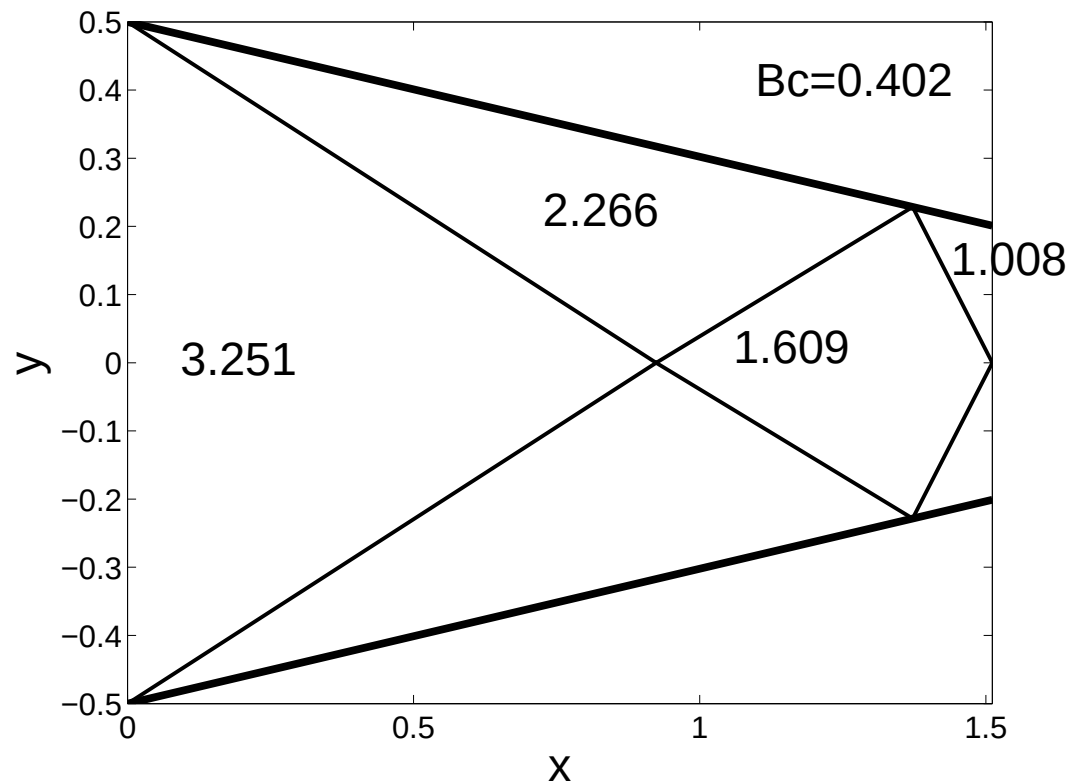


Fig. 6. Sketch of angles θ and Froude numbers F_{2m} , F_{2m+1} .

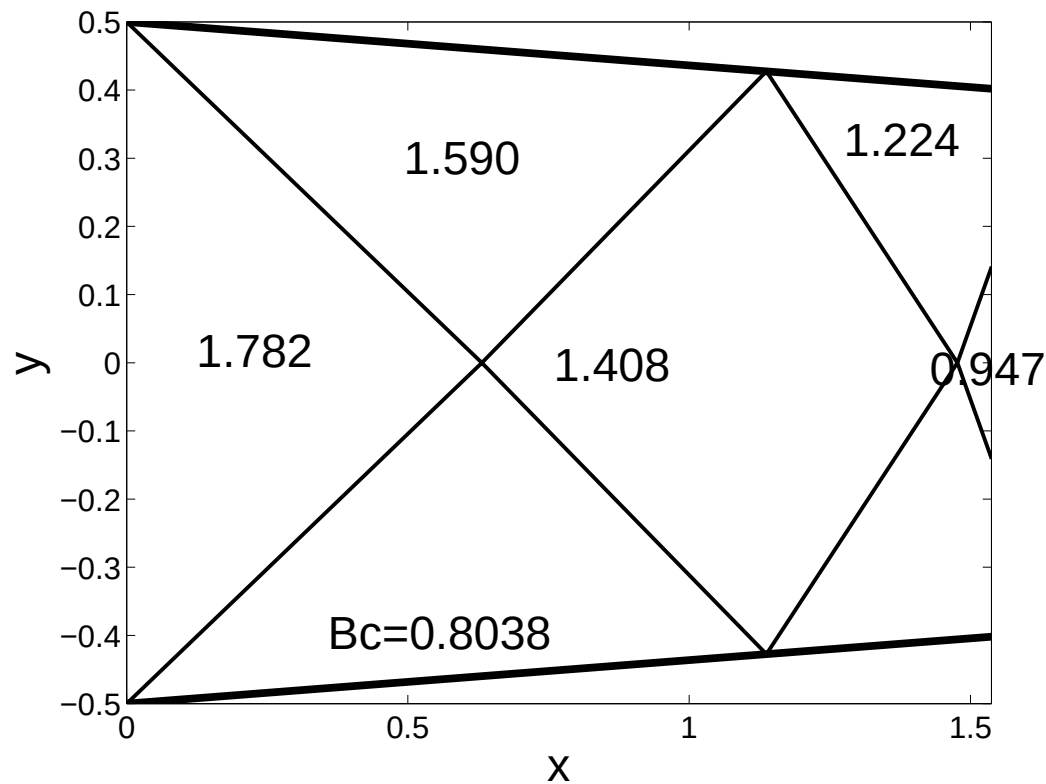
...2D theory

- when the downstream angles θ_{2m+1} or θ_{2m+2} can be calculated for Froude numbers above a critical F_0 , or



...2D theory

- Froude number of last polygon entirely fitting within contraction lies above unity for certain critical F_0 at inflow:



- Demarcation: supercritical solutions and upstream moving jumps determined with 1D/2D hydraulic theory, and simulations.

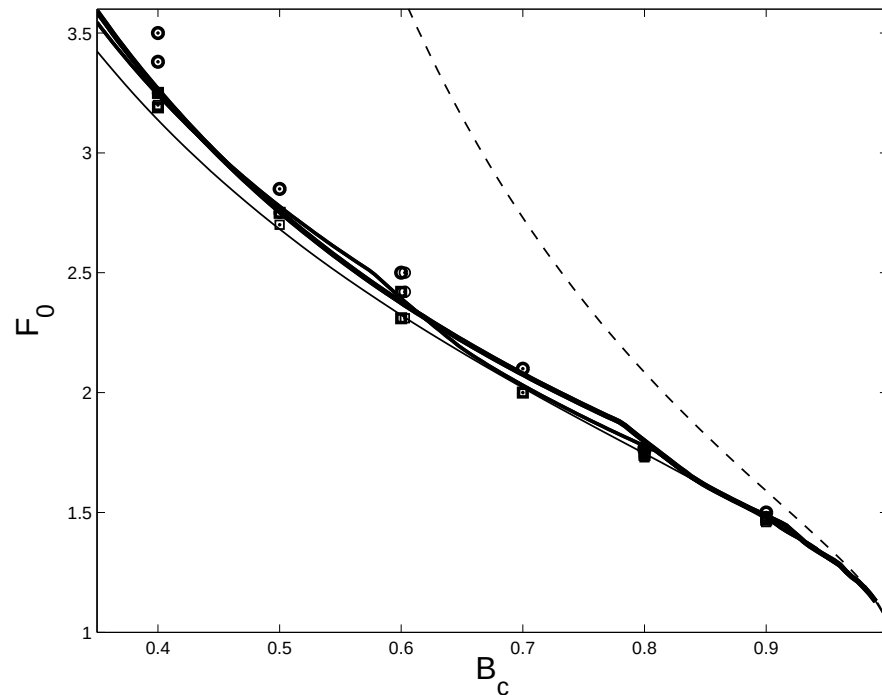
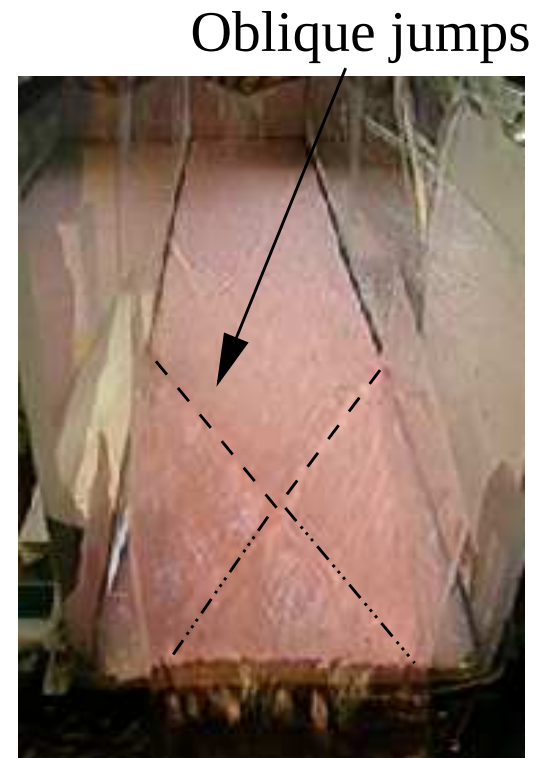
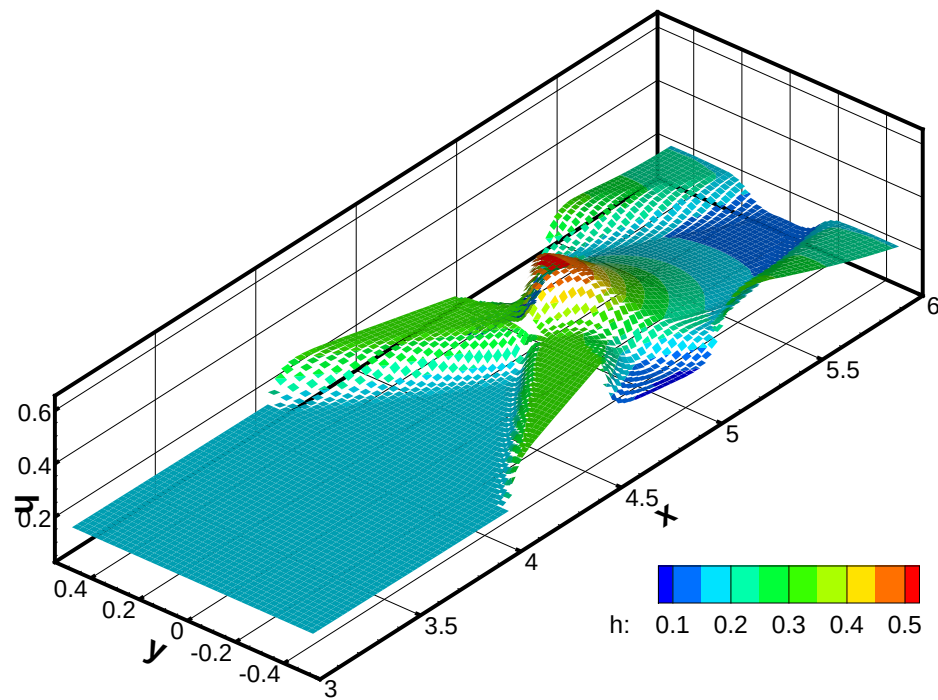
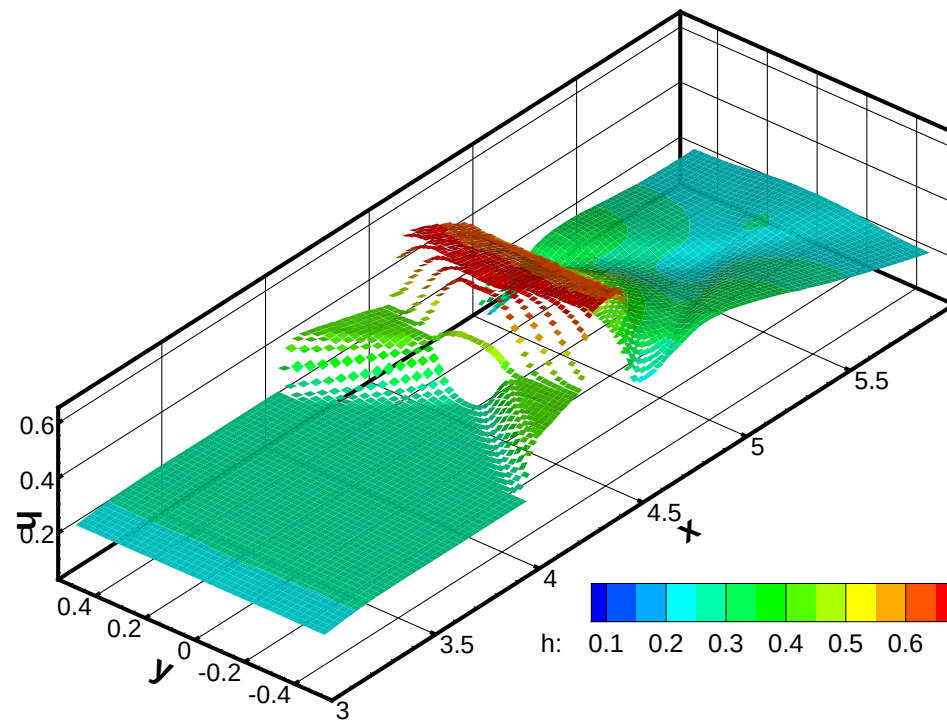


Fig. 7. 1D theory (thin solid); 2D theory: paddle lengths $L = 0.305m$ (thicker) and $L = 0.465m$ (thickest); and, simulations $L = 0.305m$ ($\circ/\square$) and $L = 0.465m$ ($\circ/\square$ with dot in center).

5. Slurry Flows



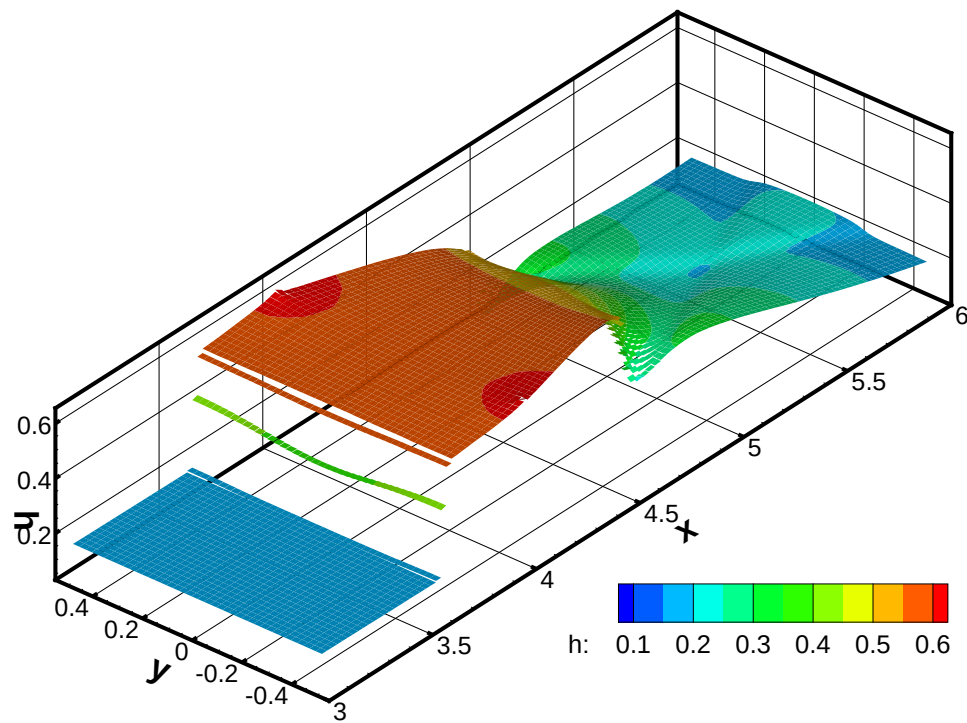
Depth-averaged 2-phase model



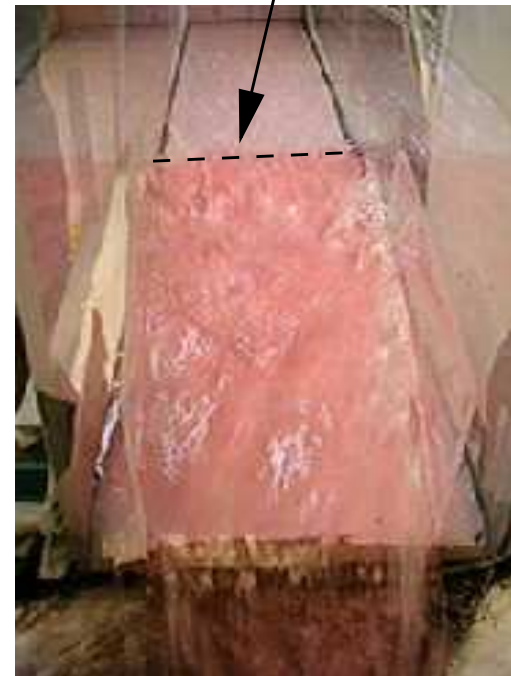
Hydraulic jump



Discontinuous Galerkin FEM

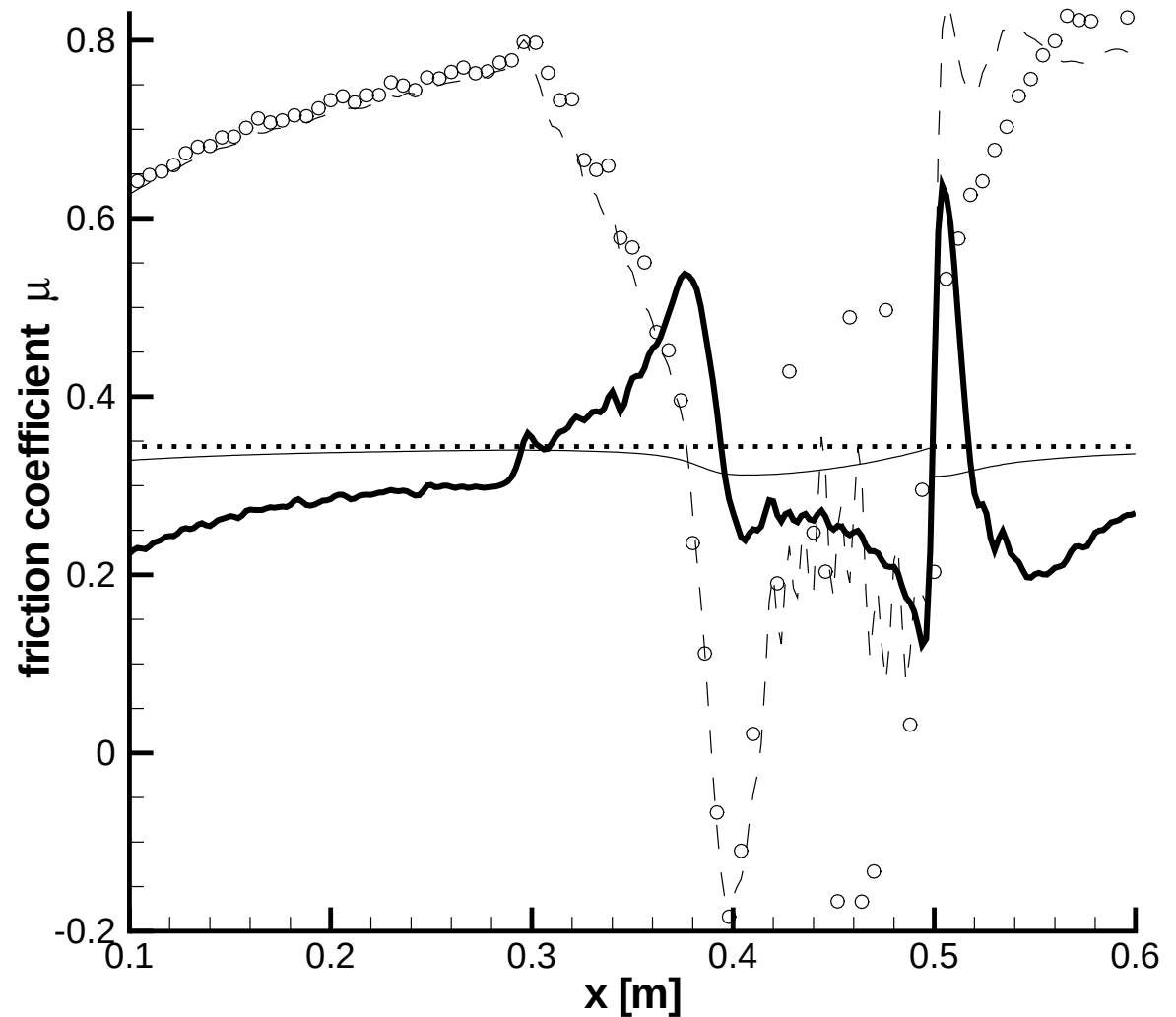


Hydraulic jump



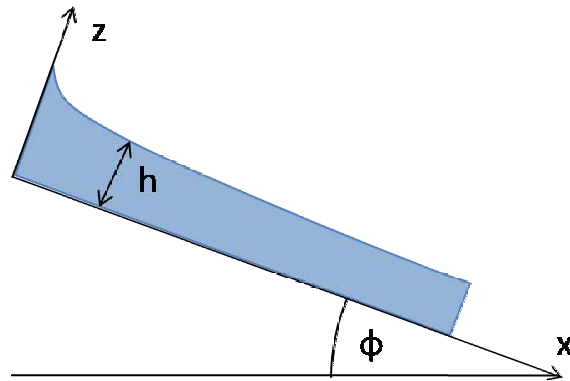
Heterogeneous Multiscale Modeling: granular flows

- **Why?**
 Rheology on
 smooth chutes
 incorrect:
 actual μ (thick solid);
 Cst. Coulomb
 friction (dotted);
 Lun et al. (1984)
 1D (dashed)
 & 2D (circles);
 Savage & Hutter
 (thin solid).



Heterogeneous Multiscale Modeling

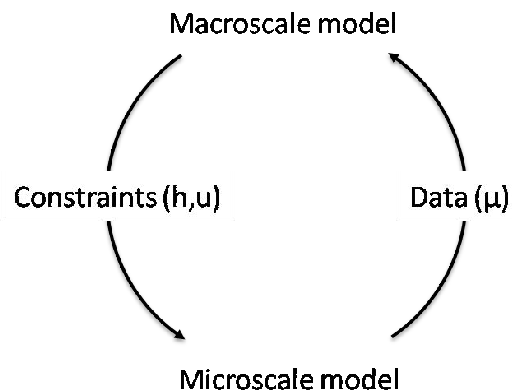
Granular chute flow



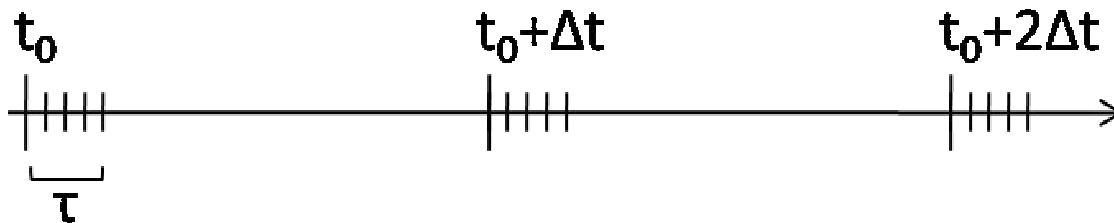
- Macro-model: Discontinuous Galerkin discretization of Shallow Water Equations with unknown basal friction coefficient μ .
- Micro-model: Molecular Dynamics simulation of granular flow around quadrature points constrained by macro-model.
- Test case: constitutive equation $\mu(h, \vec{u})$ known for flow over rough chutes (Pouliquen flow rule).

Heterogeneous Multiscale Modeling

- Coupling: Fix macro-variables while evolving micro model short time, then use data micro model to advance macro-model.



- Time-scale separation: Micro-variables relax in short time.



6. Conclusions

- *Granular* and *hydraulic* experiments and 1D theory with turbulent friction *match quite well*.
- 1D theory demarcates 2D supercritical flow boundary *surprisingly well*, for inviscid bulk flow.
- Discrete particle model simulates granular flows well: analysis.
- *Depth-averaged two-phase slurry flow simulation* recovers first experiments.
- Extension to 2D hydraulic and morphodynamic theory of oblique jumps in progress.
- Advanced experiments *fluid-particle slurry flows* in progress.
- Heterogeneous Multiscale Modeling (Thornton, Weinhart)

References

- Vreman, Al-Tarazi, Kuipers, Van Sint Annaland, & B. 2007: Supercritical shallow granular flow through a contraction: experiment, theory and simulation. *J. Fluid Mech.* 578.
- Akers & B., 2008: Hydraulic flow through a channel contraction: multiple steady states. *Phys. Fluids* 20.
- Tassi, Rhebergen, Vionnet, & B. 2008: A discontinuous Galerkin finite element model for morphological evolution under shallow flows. *Comp. Meth. Appl. Mech. Eng.* 197.
- Rhebergen, B. and Van der Vegt 2008: Discontinuous Galerkin finite element methods for hyperbolic nonconservative partial differential equations. *J. Comp. Phys.* 227.
- Pesch, Bell, Sollie, Ambati, B., & Van der Vegt 2007: hpGEM- A software framework for Discontinuous Galerkin finite element methods, *ACM Trans. Soft.* 33.
- www.math.utwente.nl/~bokhoveo/ & <http://eprints.eemcs.utwente.nl/>